Ritvik Sharma

+1-6509247092 | rsharma3@stanford.edu | Personal Website | LinkedIn | Google Scholar

Research Interests

Domain-specific compilers for Quantum Computing, Quantum Simulation and Sparse Tensor Algebra

Education

Stanford University 2021-
PhD in Electrical Engineering 4.08/4.0
Advisors: Prof. Sara Achour and Prof. Mark Horowitz

Indian Institute of Technology (IIT) Delhi 2017-2021
B.Tech. in Electrical Engineering 9.704/10.000

Experience

Software Engineering Intern at Google June'26-September'26
- Developed a new user-directed scheduling language and an agentic auto-scheduling framework for the MLIR-based Darwinn compiler stack for Edge-TPUs at Google, improving performance for Gemini and Vision ML models

Software Engineering Intern at Google June'25-September'25
- Developed KV cache pruning algorithms and system-level optimizations to enable infinite-context LLMs for on-device inference in the Litert-LM framework with the ODML Core Team at Google

Research Intern at IonQ May'22-Aug.'22
- Designed novel early fault-tolerant gate implementations for Bacon-Shor Quantum Error Correcting codes
- The low-cost implementations retain error-correction guarantees for near-term quantum computers

Publications

PLDI 2026: *CoTenN: Constrained Optimization with Tensor Networks.* **Paper**, **Code**
Ritvik Sharma, C. Peng, S. Dangwal, S. Achour

ASPLOS 2026: *Fuseflow: A Fusion-Centric Compilation Framework for Sparse Deep Learning on Streaming Dataflow.* **Paper**
R. Lacouture, O. Hsu, N. Zhang, **Ritvik Sharma**, M. Siracusa, F. Kjolstad, and K. Olukotun

Micro 2025: *A Probabilistic Perspective on Tiling Sparse Tensor Algebra.* **Paper**, **Code**
Ritvik Sharma, Z. Y. Xue, N. Zhang, R. Lacouture, F. Kjolstad, S. Achour, and M. Horowitz

PLDI 2025: *Optimizing Ancilla-Based Quantum Circuits with SPARE.* **Paper**, **Code**
Ritvik Sharma, S. Achour

PLDI 2024: *Compilation of Qubit Circuits to Optimized Qutrit Circuits* **Paper**, **Code**
Ritvik Sharma, S. Achour

ASPLOS 2023: *The Sparse Abstract Machine* **Paper**
O. Hsu, M. Strange, **Ritvik Sharma**, J. Won, K. Olukotun, J. Emer, M. Horowitz, F. Kjolstad

TECS 2022: *CODEBench: Neural Architecture and Hardware Accelerator Co-Design Framework* **Paper**
S. Tuli, C.-H. Li, **Ritvik Sharma**, N. Jha

QIP 2022: *An Algorithm for Fast Supervised Learning in Variational Circuits through Simultaneous Processing of Multiple Samples* **Paper**
S. Dangwal, **Ritvik Sharma**, D. Bhowmik

ISCAS 2021: *A Crossbar Array of Analog-Digital-Hybrid Volatile Memory Synapse Cells for Energy-Efficient On-Chip Learning* **Paper**
J. Sharda, **Ritvik Sharma**, D. Bhowmik

Selected Research Projects

DSL and Compiler for Tensor-Network (TN) based Quantum Simulator Jan.'25-Present
Under Prof Sara Achour, NCS Lab, Stanford University
- Designed a language to express different Hamiltonian Simulations as Constrained Optimization Problems, solved under a low-rank approximation using Tensor Networks with a Matrix Product States/Operators (MPSs/MPOs).
- Compiles constraints to sparse-tensors, enabling faster Tensor Network algorithms (published at PLDI 2026).

Rewrite-based Compilers for Quantum Circuits Feb.'23-Present

Under Prof Sara Achour, NCS Lab, Stanford University

- Developed a quantum circuit compiler for control-flow-based high-level quantum computing languages (SPARE)
- Circuit optimization using rewrites that reduce the generated quantum circuit size by 2-7×. Published in PLDI 2025.
- Developed the DARE compiler to circuits that use qudit higher-energy states.
- DARE verified circuit rewrites to reduce the circuit size by up to 2.1×. Published in PLDI 2024.

Sparse Tensor Algebra Tiling Optimization Sept.'23-Sept.'24

Under Prof Mark Horowitz, Stanford University

- Developed a performance model for sparse tensor algebra, which models data distribution probabilistically
- Identify and collect easy-to-access high-level statistics from compressed sparse data formats
- Implemented a static sparse tiling optimization tool (D2T2) based on the probabilistic traffic model, reducing traffic by 1-2 order of magnitude. Published in Micro 2025.

Architecture Simulations for Sparse Tensor Algebra accelerators March'22-March.'23

Under Prof Mark Horowitz, AHA Project, Stanford University

- Built a cycle-approximate architecture simulator for Sparse Abstract Machine (published at ASPLOS 2023)
- Developing a calibration framework for architecture-level simulations against performance of taped-out hardware for cross-platform performance validation

Undergraduate Research Assistant at IIT Delhi Sept.'18-May'21

- Designed CMOS-analog on-chip-memory based accelerators using RRAMs and other non-volatile memories
- Designed a novel Quantum Machine Learning algorithm that encodes and processes multiple data samples together providing exponential speedups for model training in Variational Classifiers

Undergraduate Research Intern at UCSD May-August 2019

- Tested a monolithic CMOS-RRAM compute-in-memory accelerator for ML inference and training
- Tested a hybrid 10-bit resolution asynchronous ADC chip with 5-bits of level crossing and 5-bits with flash ADC

Undergraduate Research Assistant at Princeton University May'20-Aug'21

- Created an accelerator-CNN co-design tool that explores a broad range of CNNs sampled from a family of model search space for Accelerator design
- Implemented RTL code for implementing the accelerator design and memory simulations

Achievements

- Awarded **Stanford Graduate Fellowship (SGF)** by Stanford University
- **Department rank 2** in a batch of 85+ students with GPA of 9.704/10 at IIT Delhi
- Awarded **semester merit award** for being in the top 7% of the class in 6 out of 8 semesters at IIT Delhi

Relevant Courses

Computer Science: Compilers for Domain-Specific Languages, Compilers, Parallel Processors Beyond Multi-Core Processing, Reinforcement Learning, Quantum Computing, Computer Architecture for HPC, Computer Architecture, Advanced Machine Learning, Data Structures, Computational Complexity

Electrical Engineering: VLSI Systems, Design Project in VLSI, Image Systems Engineering, Coding Theory, Digital Image Processing, Neuromorphic Engineering, Communication Engineering

Physics: Quantum Computing, Theory of General Relativity, Electromagnetics, Quantum Mechanics

Technical Skills

Languages: Python, C++, C, Julia, MATLAB, Java, Verilog, Assembly (x86-64, RISC and MIPS)
Libraries and packages: Scipy, Tensorflow, ITensor, ITensorsMPS, Qiskit, Cirq, Taco Sparse Compiler, PyTorch, Eigen
Frameworks and Tools: HLS (catapult and Vivado), Cadence and Synopsis EDA Tools

Teaching Experience

Teaching Assistant at IIT Delhi, for course EE 205, Signals and Systems for Fall and Spring Semester 2020-21.